# Aleksei Lian

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## Education

**York University** — Toronto, ON
*Bachelor of Science in Computer Science* — *Expected Dec. 2027*
*Transferred from Carleton University* — *Sep. 2023 – Dec. 2024*

## Experience

**Software Engineer (Co-Founder)** — May 2026 – Present
*Arzan (China-to-Kyrgyzstan E-Commerce, Pre-Launch)* — *Remote*

- Built a 90+ endpoint FastAPI/PostgreSQL API, React staff console, and React Native app as the sole engineer.
- Fixed a race condition that let two operators buy the same order on Taobao, using PostgreSQL row locks.
- Built job queues with FOR UPDATE SKIP LOCKED, so parallel workers never claim the same row.
- Made checkout idempotent with a unique database key, so a retried request never places a second order.
- Enforced money, quantity, and status rules with 50 PostgreSQL CHECK constraints that reject invalid data.
- Set up GitHub Actions CI that runs the 2,300+ test backend suite on both SQLite and PostgreSQL.

**Backend Engineer (Co-Founder)** — Apr. 2026 – Aug. 2026
*Tanish (Dating App for Central Asia, iOS Beta)* — *Remote*

- Shipped an iOS app to a TestFlight beta: FastAPI/PostgreSQL backend, WebSocket chat, React Native client.
- Designed JWT auth with 15-minute access tokens and rotating refresh tokens that revoke the chain on reuse.
- Verified real users with AWS Rekognition face liveness, using STS credentials that can only start the stream.
- Closed a hole where anyone with the project's public key could read and write user tables through Supabase's auto-generated API, by enabling deny-by-default row-level security.
- Fixed moderation jobs that never retried (arq re-queues only on its own Retry signal) by adding real retries and an hourly sweep that re-screens missed chat and voice content.

## Projects

**Temir AI** - AI Quoting Startup (Co-Founder) | *JavaScript, Anthropic API, PostgreSQL* — Aug. 2026 – Present

- Built the product for our pilot customer, a metal supplier, where **4 sales managers at 2 warehouses** used it to turn orders sent as chat text, PDF, spreadsheet, or photo into priced quotes.
- Kept the LLM from setting quote prices: order parsing returns line items, and unit prices come from the catalog.
- Capped API spend per user and company-wide by reserving budget under a PostgreSQL advisory lock.

**Mendy** - AI Advice Companion | *Python, FastAPI, PostgreSQL, OpenAI API* — Jul. 2026

- Built the backend for an AI advice companion with long-term memory, proactive follow-ups, and crisis escalation.
- Caught a prompt edit that lowered a crisis message's risk tier, using a 40-message safety test set that must flag all 12 crisis cases and is re-run by hand after every safety-prompt change.

**URL Shortener** | *Python, FastAPI, asyncpg, PostgreSQL, Docker, GCP* — Feb. 2026 – Mar. 2026

- Built an async URL shortener where two simultaneous requests can't get the same short code: inserts retry on a primary-key conflict instead of check-then-insert.
- Deployed it to Google Cloud Run with Cloud SQL via Cloud Build, with GitHub Actions CI running 38 tests, 23 against a live PostgreSQL container.

**Chess Engine** | *Python, UCI* — Feb. 2026

- Built a UCI chess engine that plays timed games, using alpha-beta search and a transposition table.

## Technical Skills

**Languages**: Python, SQL, TypeScript, JavaScript, Java, C
**Frameworks**: FastAPI, SQLAlchemy, Pydantic, Alembic, pytest, React, React Native, LLM APIs (OpenAI, Anthropic)
**Data & Cloud**: PostgreSQL (transactions, locking), Redis, SQLite, AWS, GCP (Cloud Run, Cloud SQL), OpenTofu
**Developer Tools**: Git, GitHub Actions (CI/CD), Docker, Linux, REST APIs, Claude Code (AI-assisted development)